SMART CITY · MULTI-AGENT SYSTEMS · PROJECT 01

Disaster Response Coordination

Building an Autonomous, Multi-Agent Decision Engine for Urban Crises

> **THE 90-SECOND MISSION**
>
> *"It's 2:00 AM. A river gauge spikes. Water rises rapidly toward three densely populated neighborhoods. You have one ambulance fleet, a single shelter with 40 beds, and zero time to waste. Your mission: build the autonomous, multi-agent AI brain that turns raw sensor chaos into a life-saving rescue plan — before a human even picks up the phone."*

The 6-Stage Journey at a Glance

Stage	Mission Objective	Delivery Schedule
01. Machine Learning	Spot the Pattern: Turn raw sensor chaos into a live zone risk score.	Day 2 Build → Level-01 Check (Day 4)
02. Deep Learning	Develop Instincts: See through floodwaters using live camera feeds & satellite imagery.	Day 3 Build → Level-01 Check (Day 4)
03. Natural Language Processing	Decode the Panic: Parse unstructured dispatcher logs, 911 calls & emergency texts.	Day 5 Build → Level-02 Check (Day 7)
04. Small Language Model	Brief the Commander: Deliver fast, 5-second, voice-ready field summaries offline.	Day 6 Build → Level-02 Check (Day 7)
05. Generative AI	Stress-Test Reality: Simulate impossible compound disaster scenarios.	Day 8 Build → Checkpoint (Day 10)
06. Agentic AI	Take the Wheel: Autonomous reasoning, resource planning, and unit dispatch.	Day 9 Build → Checkpoint (Day 10)

STAGE 01: Machine Learning

Teach a model to see the pattern first

Mission: Before an AI agent can save a city, it needs to understand the threat level. Turn continuous streams of sensor data, weather patterns, and 911 calls into an instant risk score for every city zone.

Your Data:

- **River & Rainfall Gauges:** 72-hour rolling window per district
- **Historical Flood Logs:** Historical incident outcomes at matching gauge levels
- **Emergency Calls:** Real-time 911 call volumes grouped by neighborhood

- **Infrastructure Records:** Live road and bridge closure logs

What Success Looks Like:

- **Zone Risk Score Model:** Real-time triage classifier (Low / Moderate / Severe)
- **Feature Leaderboard:** Data-driven breakdown of leading risk predictors
- **Field Briefing Sheet:** 1-page non-technical responder reference guide

Squad Roles & Responsibilities:

Role	Core Responsibilities
Data Engineer	Stitch gauge, weather, and 911 logs into a clean, timestamped master dataset.
EDA Engineer	Plot leading indicators; hunt down and eliminate historical false alarms.
ML Engineer	Train and tune the core classifier scoring zone risk in real time.
Evaluation Engineer	Stress-test the model against unseen past disasters to flag overconfidence.
Integration Engineer	Wrap the model in an ultra-fast API for real-time dashboard consumption.

> **Team Huddle:** Before writing code, agree as a team on the exact numerical definition of "Severe". Get this wrong, and every downstream agent inherits the flaw.

Schedule: Day 2 Build → Level-01 Check (Day 4)

STAGE 02: Deep Learning

Go from patterns to instincts

Mission: Numbers alone miss crucial context. Upgrade your system to visually analyze street cameras, drone feeds, and temporal sensor sequences to spot disasters unfolding in real time.

Your Data:

- **Visual Feeds:** Drone and CCTV frames (clear, partially blocked, and flooded streets)
- **Satellite Time-Series:** High-resolution satellite imagery across affected districts
- **Temporal Streams:** Sequential gauge sensor readings treated as time-series data

What Success Looks Like:

- **Computer Vision CNN:** Automatically flags submerged roadways and blocked access routes
- **Sequence Model (LSTM/Transformer):** Projects water level trends 3 hours into the future
- **Visual Benchmark Reel:** Side-by-side comparison of DL visual calls versus ML baselines

Squad Roles & Responsibilities:

Role	Core Responsibilities
Data Engineer	Label street footage; perform data augmentation for night and heavy rain conditions.
EDA Engineer	Inspect model saliency maps to verify it focuses on floodwater, not dark shadows.
DL Engineer	Build and train the CNN vision network and LSTM/Transformer sequence model.
Evaluation Engineer	Refine confusion matrices until the model stops mistaking wet asphalt for deep floods.
Integration Engineer	Fuse vision and time-series scores into a single unified alert API.

> **Team Huddle:** Inspect 5 borderline "not-quite-flooded" images together. If the team can't agree on the classification, that's your real benchmark—train for these edge cases.

Schedule: Day 3 Build → Level-01 Check (Day 4)

STAGE 03: Natural Language Processing (NLP)

Give the system a sense of language

Mission: During a crisis, critical details hide in unstructured text—911 transcripts, police radio notes, and social media calls for help. Teach the system to extract structure from human communication.

Your Data:

- **Dispatcher Logs:** Historical incident text logs and emergency call notes
- **Safety SOPs:** Public safety bulletins and Standard Operating Procedures
- **Social Feeds:** Simulated social media distress messages and emergency posts

What Success Looks Like:

- **Urgency Text Classifier:** Categorizes incoming messages by urgency level and hazard type
- **Named-Entity Recognizer (NER):** Automatically extracts Location, Resource Needed, and Headcount
- **Misinterpretation Audit Log:** Documented breakdown of challenging misreads and resolutions

Squad Roles & Responsibilities:

Role	Core Responsibilities
Data Engineer	Gather, sanitize, and annotate raw text logs for urgency classification and NER.
EDA Engineer	Analyze vocabulary distributions and phrasing patterns across urgent vs. routine reports.
NLP Engineer	Construct the text classification pipeline and entity extraction NER models.
Evaluation Engineer	Benchmark precision/recall on held-out test logs; analyze edge-case failures.
Integration Engineer	Wire the pipeline into a live intake UI where dispatchers instantly see tagged entities.

> **Team Huddle:** Read 3 real disaster tweets out loud. Notice how urgency is often implied rather than explicitly stated. That is the exact gap your NLP model must bridge.

Schedule: Day 5 Build → Level-02 Check (Day 7)

STAGE 04: Small Language Model (SLM)

Make it fast, local, and conversational

Mission: An incident commander in the field cannot read lengthy incident logs—they need a 5-second voice briefing. Fine-tune a compact, local model capable of generating lightning-fast summaries on edge devices.

Your Data:

- **Curated Summaries:** High-quality report-to-summary training pairs generated from Stage 03
- **Domain Dictionary:** Evacuation terms, resource codes, and tactical radio shorthand

What Success Looks Like:

- **Fine-Tuned Local SLM:** Condenses dense multi-page reports into 2 crisp, actionable sentences
- **Performance Benchmarks:** Latency and accuracy comparisons against massive cloud models
- **Offline Laptop Demo:** Fully operational edge application running locally with zero cloud GPU dependency

Squad Roles & Responsibilities:

Role	Core Responsibilities
Data Engineer	Curate and format report-summary pairs into a clean fine-tuning dataset.
EDA Engineer	Audit token distribution to ensure specialized radio codes are retained.
SLM Engineer	Fine-tune the small language model using parameter-efficient tuning techniques.
Evaluation Engineer	Benchmark perplexity, summary fidelity, and inference latency under stress.
Integration Engineer	Package the SLM into an offline tactical briefing mobile/desktop app.

> **Team Huddle:** Time yourselves reading a full incident log versus reading the SLM's 2-line summary. If the time savings isn't dramatic (>80% reduction), refine the summary parameters.

Schedule: Day 6 Build → Level-02 Check (Day 7)

STAGE 05: Generative AI

Teach it to imagine what hasn't happened yet

Mission: Major disasters are fortunately rare, resulting in sparse training data. Utilize Generative AI to synthesize complex, high-impact disaster scenarios to battle-test the entire pipeline before real deployment.

Your Data:

- **Historical Disaster Seeds:** Statistical distributions and patterns from past natural disasters
- **Synthetic Event Generators:** Multi-point generated sensor values paired with synthetic emergency text logs

What Success Looks Like:

- **Scenario Generator:** Produces highly realistic, multi-zone compound disaster scenarios
- **Stress-Test Evaluation:** Comprehensive performance report evaluating system response across 20 synthetic events
- **The "Wildcard" Challenge:** A custom team-invented scenario presented and defended live

Squad Roles & Responsibilities:

Role	Core Responsibilities
Data Engineer	Assemble reference baseline datasets from real historical disaster distributions.
EDA / Prompt Eng.	Identify blind spots in historical data and craft prompts for extreme edge scenarios.
GenAI Engineer	Construct the generative scenario pipeline (GAN/VAE or LLM-based synthetic generation).

Evaluation Engineer	Audit generated scenarios to verify they realistically stress system logic.
Integration Engineer	Integrate synthetic scenario output back into the testing dashboard for continuous evaluation.

> **Team Huddle:** Invent one scenario no one on the team has considered yet — e.g., a flash flood combined with a total municipal power outage. Run it through your pipeline as your capstone story.

Schedule: Day 8 Build → Checkpoint (Day 10)

STAGE 06: Agentic AI

Let it act on its own judgement

Mission: This is where all components converge. Build an autonomous agent that synthesizes risk scores, vision alerts, and text intelligence to formulate rescue plans, allocate resources, and dispatch units independently.

Your Data:

- **Live Sensor Intelligence:** Real-time feeds from ML risk scores, DL vision alerts, and SLM briefings
- **Resource Inventory:** Live database of available ambulances, rescue personnel, and shelter beds
- **Response Playbooks:** Standard Operating Procedures serving as the agent's core knowledge base

What Success Looks Like:

- **Autonomous Dispatch Engine:** Generates multi-zone allocation plans for active simulated disasters
- **Transparent Reasoning Trace:** Step-by-step decision log explaining the logic behind every dispatch
- **Human Override Control:** Tested emergency pause/override interface for tactical commanders

Squad Roles & Responsibilities:

Role	Core Responsibilities
Knowledge Engineer	Construct the SOP knowledge base and interface tools for shelter registries & fleet APIs.
Workflow Engineer	Deconstruct macro goals ("Evacuate Zone 7") into concrete, tool-executable subtasks.
Agent Engineer	Design the core reasoning loop, tool invocation parameters, and decision logic.
Evaluation Engineer	Execute simulated incident runs; catalog task success rates and decision edge failures.
Integration Engineer	Build the Orchestration Command Center UI featuring live execution monitoring and manual override.

> **Team Huddle:** Deliberately present the agent with an impossible dilemma: two zones flooding simultaneously with only one ambulance available. Observing how it prioritizes trade-offs is crucial.

Schedule: Day 9 Build → Checkpoint (Day 10)

Expected Deliverables & Study Materials

Expected Deliverables:

- **Stage 01 (ML):** Real-Time Zone Risk Score API, Feature Leaderboard & Importance Analysis, 1-Page Responder Field Briefing Sheet.
- **Stage 02 (DL):** Computer Vision Water Level & Blockage Classifier, 3-Hour Predictive Water Level Sequence Model (LSTM/Transformer), Visual Benchmark Comparison Reel.
- **Stage 03 (NLP):** Urgency Text Classifier, Named-Entity Recognition (NER) Pipeline for Hazard/Location Extraction, Misinterpretation Audit Log.
- **Stage 04 (SLM):** Fine-Tuned Local Small Language Model for Tactical Summaries, Inference Latency & Accuracy Benchmark Report, Offline Edge Demonstration App.
- **Stage 05 (Gen AI):** Multi-Zone Synthetic Scenario Generator, 20-Scenario Stress-Test Evaluation Report, "Wildcard" Compound Disaster Capstone Scenario & Live Defense.
- **Stage 06 (Agentic AI):** Autonomous Dispatch Engine with Multi-Zone Resource Allocation, Transparent Decision & Reasoning Audit Logs, Tactical Command Center UI with Manual Override Controls.
- **Final Capstone:** Fully integrated end-to-end multi-agent command system demo, unified codebase repository, and complete technical architecture documentation.

Study Materials & Interactive Learning Web App:

Study material for this project is hosted as an interactive web application designed for learning basic concepts and exploring foundational topics across all 6 stages. The access link to the web app will be shared later.

- **Machine Learning & Tabular Triage:** Classification fundamentals, decision trees, ensembles, and handling imbalanced crisis dataset distributions.
- **Deep Learning & Computer Vision:** Convolutional neural networks (CNNs), aerial imagery feature extraction, and temporal sequence prediction using LSTMs and Transformers.
- **Natural Language Processing & NER:** Text classification pipelines, tokenization, and custom Named-Entity Recognition (NER) for extracting emergency details.
- **Small Language Models & Edge AI:** Compact model architectures, parameter-efficient fine-tuning (PEFT/LoRA), and local CPU/GPU offline deployment techniques.
- **Generative AI & Scenario Simulation:** Synthetic data generation, edge-case generation for rare disaster scenarios, and automated red-teaming AI evaluation.
- **Agentic AI & Multi-Agent Systems:** Autonomous tool invocation, reasoning loops (ReAct pattern), multi-agent negotiation, and human-in-the-loop safety controls.

Days 1–4	Days 5–7	Days 8–10	Day 11-12
ML + DL Level-01 Milestone	NLP + SLM Level-02 Milestone	Gen AI + Agentic AI System Checkpoint	Full System Capstone Launch